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Date: 17.08.2026

Review- 1

Students should download research papers relevant to their chosen concept/title from **Scopus** and **Web of Science (WoS)** indexed databases. Only downloadable, full-text **journal papers** published **from 2021** and directly related to your proposed concept should be selected.

Ensure that a valid, working downloadable link is available for each paper, and that the dataset, methodology, and limitations discussed in each paper are clearly identifiable, as these will be tabulated below.

S.No	Name of the paper and Authors	Year of Publication	Downloadable Link	Journal Name/Scopus or Web of Science	Dataset Used	Methodology Used	Limitations
1	Blind Spot Detection Radar System Design for Safe Driving of Smart Vehicles — Wantae Kim, Heejin Yang, Jinhong Kim	2023	https://doi.org/10.3390/app13106147	Applied Sciences (MDPI) — Scopus & Web of Science (SCIE)	No public benchmark dataset reported. Simulation data and custom real-world vehicle tests using the developed BSD radar system.	FMCW radar; ADC; 1D/2D FFT; range-Doppler mapping; peak detection; angle estimation; target sorting and tracking; α-β / α-β-γ tracking filters; custom antenna and modem design; vehicle-level testing.	Complex driving environments make multi-target prioritization difficult. Equal tracking time for all targets can cause tracking failures. The paper identifies AI-based target prioritization as future work.
2	Blind-Spot Monitoring System using Lidar — Kuldeep S. Pawar, Shivanand N. Teli, Prasad Shetye, Saukshit Shetty, Vedant Satam, Atul Sahani	2022	https://doi.org/10.1007/s40032-022-00856-2	Journal of The Institution of Engineers (India): Series C — Scopus	No public benchmark dataset reported. Results were obtained from on-road tests in regular traffic using the developed LiDAR prototype.	Retrofittable wireless BSD using TF-Luna LiDAR, Arduino UNO, HC-05 Bluetooth and BLiM mobile application. Distance-based detection with LED and buzzer alerts. Real-time on-road testing.	Additional sensors are needed to cover different blind-spot zones. The approach is mainly distance-based and does not provide the richer range/velocity/angle information available from FMCW radar. Testing was limited to the developed prototype and reported scenarios.
3	A Multi-Scale Mask Convolution-Based Blind-Spot Network for Hyperspectral Anomaly Detection — Zhiwei Yang, Rui Zhao, Xiangchao Meng, Gang Yang, Weiwei Sun, Shenfu Zhang, Jinghui Li	2024	https://doi.org/10.3390/rs16163036	Remote Sensing (MDPI) — Scopus & Web of Science (SCIE)	Four real hyperspectral datasets: Pavia, Gainesville, San Diego-1 and Gulfport. Sensors include ROSIS-03 and AVIRIS.	Multi-scale blind-spot convolution; dynamic fusion; spatial-spectral joint module; background feature attention; pixel-shuffle down-sampling; spatial-spectral screening; self-supervised hyperspectral anomaly detection; comparison with nine state-of-the-art methods.	This paper addresses hyperspectral anomaly detection rather than automotive blind-spot detection. Its datasets are hyperspectral images, not automotive radar/LiDAR data, so direct transfer to vehicle BSD is not established.

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Based on the datasets, methodologies, and limitations identified from the reviewed literature above, students should clearly articulate how their **proposed system** differs from and improves upon the existing approaches. This justification should explicitly address the **novelty** of the proposed work by describing which limitations of the existing methods/datasets are being overcome, what new dataset(s) or methodology are being introduced or adapted, and why the proposed approach is expected to perform better or offer a meaningful contribution over the reviewed papers. **(POINT WISE)**

Proposed System Novelty and Improvement (Point Wise)

1. Use FMCW radar as the primary sensing technology to obtain range, relative velocity and angle information for moving vehicles.
2. Add AI-based threat assessment to address the target-prioritization limitation identified in the FMCW radar paper.
3. Adapt a public automotive radar dataset such as RadarScenes and generate left/right blind-spot labels from object position, velocity and tracking information.
4. Implement multi-target prioritization using distance, relative velocity, position and estimated collision risk.
5. Include Time-to-Collision (TTC) as an additional feature so approaching vehicles can be assigned a higher risk level.
6. Extend the system with camera-radar sensor fusion: radar provides range/velocity/angle while the camera provides visual object classification.
7. Monitor both left and right blind-spot zones, addressing the LiDAR paper's observation that additional sensors may be needed for wider coverage.
8. Provide real-time visual/audio driver warnings based on the calculated risk level.
9. Maintain a clear automotive focus: the third paper's 'blind-spot network' is for hyperspectral anomaly detection and is not an automotive BSD system.
10. Expected contribution: combine FMCW radar sensing, target tracking, blind-spot zoning and AI-based threat prioritization to improve reliability in multi-target driving situations.

Note: The three papers are from journals indexed in Scopus. Applied Sciences and Remote Sensing are also indexed in Web of Science SCIE. DOI links above provide the paper access pages.